MODULE 1: Introduction and Database Architecture

Section A

1. Define a database.
2. What is a DBMS?
3. List the three views/levels of data in a database system.
4. What is data abstraction?
5. Define a data model.
6. Name any two types of data models.
7. What is the physical level in database architecture?
8. What is the logical level of data abstraction?
9. What is the view level in DBMS?
10. Define schema.
11. What is an instance of a database?
12. What is a relational database?
13. Define a relation in the relational model.
14. What is a tuple?
15. What is an attribute domain?
16. Define primary key.
17. What is a candidate key?
18. What is a foreign key?
19. Name any two operators of relational algebra.
20. What is a super key?

Section B

1. Differentiate between DDL and DML.
2. Explain the three levels of database architecture briefly.
3. What is data independence? Name its types.
4. Distinguish between physical and logical data independence.
5. Define database design and explain its importance.
6. What are the components of the relational model?
7. Explain the concept of a super key with an example.
8. Differentiate between candidate key and primary key.
9. What is a foreign key? Give an example using two tables.
10. Explain the SELECT (σ) operation of relational algebra with syntax.
11. Explain the PROJECT (π) operation of relational algebra with an example.
12. Write a relational algebra expression to find employees with salary > 50000 from EMPLOYEE(EmpID, Name, Salary).
13. Explain the UNION operation in relational algebra and the conditions required for it.
14. What is the SET DIFFERENCE operation? Give an example.
15. Explain the Cartesian product operation with an example.
16. What is a JOIN operation in relational algebra?
17. Differentiate between theta join and equi join.
18. What is a natural join? Give an example.
19. Explain the DIVISION operation in relational algebra with a use case.
20. Write a relational algebra query to find names of students who scored above 90 from STUDENT(RollNo, Name, Marks).

Section C

1. Explain the three-schema architecture of DBMS with a neat diagram.
2. Discuss the different data models used in database systems with examples.
3. Explain the purpose of a database system and discuss the disadvantages of file-processing systems.

4. Describe the structure of a relational database with a suitable example table.
5. Explain the different types of keys in the relational model (primary, candidate, foreign, super, composite) with suitable examples.
6. Given relations STUDENT(RollNo, Name, DeptID) and DEPT(DeptID, DeptName), write and explain relational algebra expressions for: (a) students in the "CSE" department, (b) names of all students, (c) student and department details combined using a join.
7. Explain the fundamental operations of relational algebra with syntax and an example for each: Select, Project, Union, Set Difference, Cartesian Product.
8. Explain additional relational algebra operations — Intersection, Assignment, and Rename — with suitable examples.
9. Design a relational schema for a "Library Management System" identifying entities, attributes, and keys, and explain how relational model concepts apply to it.
10. Explain database system architecture (centralized, client-server, distributed) and discuss the role of a Database Administrator (DBA).

MODULE 2: SQL and its Application

Section A

1. What is SQL? Expand the term.
2. Expand DDL.
3. Expand DML.
4. Name any two DDL commands.
5. Name any two DML commands.
6. What is the use of the WHERE clause?
7. What is a NULL value in SQL?
8. Name any two aggregate functions in SQL.
9. What is the purpose of the GROUP BY clause?
10. What is the HAVING clause used for?
11. What is a nested query?
12. Define a view in SQL.
13. What is a transaction in a DBMS?
14. What is meant by an integrity constraint?
15. What is a foreign key constraint used for?
16. What is embedded SQL?
17. What is dynamic SQL?
18. Define a trigger in SQL.
19. What is a stored procedure?
20. What is the purpose of the GRANT command?

Section B

1. Write the basic syntax structure of an SQL SELECT query.
2. Write an SQL query to create a table STUDENT(RollNo int, Name varchar(20), Marks int).
3. Write an SQL query to insert a record into the STUDENT table.
4. Write an SQL query to display all students with Marks greater than 75.
5. Differentiate between the WHERE clause and the HAVING clause.
6. Write an SQL query using UNION to combine results of two tables with identical structure.
7. Explain NULL values and how they are handled in SQL, with an example.
8. Write an SQL query to find the average marks of students using an aggregate function.

9. Explain nested queries with a simple example.
10. Write an SQL query to update the marks of a student with RollNo = 5.
11. Write an SQL query to delete a student record with RollNo = 10.
12. Explain the difference between INNER JOIN and OUTER JOIN.
13. Write an SQL query to create a view showing student names with Marks > 80.
14. Explain integrity constraints in SQL with a suitable example.
15. What is authorization in SQL? Name two types of privileges.
16. Explain the difference between the DELETE and DROP commands.
17. Write an SQL query using GROUP BY and COUNT to count students per department.
18. What is embedded SQL? Give a brief example scenario.
19. Explain a trigger with a simple example (syntax outline only).
20. Write an SQL query to create a stored procedure that returns student details for a given RollNo.

Section C

1. Explain the basic structure of SQL queries (SELECT–FROM–WHERE) with suitable examples covering all clauses (ORDER BY, DISTINCT, etc.).
2. Given tables EMPLOYEE(EmpID, Name, DeptID, Salary) and DEPARTMENT(DeptID, DeptName), write SQL queries for: (a) employees with salary > 50000, (b) employee names with department name using a join, (c) total salary department-wise using GROUP BY, (d) employees earning more than the average salary using a nested query.
3. Explain the set operations in SQL — UNION, INTERSECT, and EXCEPT — with syntax and examples.
4. Explain the various aggregate functions in SQL (COUNT, SUM, AVG, MAX, MIN) with an example query for each.
5. Discuss views in SQL — creation, updating, and dropping a view — with advantages and suitable examples.
6. Explain the concept of transactions in SQL, including COMMIT, ROLLBACK, and the ACID properties.
7. Explain the various types of integrity constraints in SQL (Primary Key, Foreign Key, Unique, Check, Not Null) with example CREATE TABLE statements.
8. Explain embedded SQL and dynamic SQL, their differences, and typical use cases.
9. Explain triggers in SQL with syntax, types (BEFORE/AFTER, ROW/STATEMENT level), and a suitable example.
10. Explain authorization and privileges in SQL — the GRANT and REVOKE commands — with suitable examples.

MODULE 3: Database Design

Section A

1. What is an entity?
2. What is an entity set?
3. Define attribute in the ER model.
4. What is a relationship in the ER model?
5. Name the types of attributes in the ER model.
6. What is a weak entity?
7. What is a strong entity?
8. Define cardinality ratio.
9. Name the types of cardinality ratios.
10. What is a participation constraint?

11. What is meant by redundancy in ER modelling?
12. What is a functional dependency?
13. Name Armstrong's Axioms.
14. Define normalization.
15. What is 1NF?
16. What is 2NF?
17. What is 3NF?
18. What is BCNF?
19. Define multivalued dependency.
20. What is 4NF?

Section B

1. Differentiate between an entity and an entity set.
2. Explain simple and composite attributes with an example.
3. Explain single-valued and multi-valued attributes with an example.
4. What is a derived attribute? Give an example.
5. Differentiate between a strong entity and a weak entity with an example.
6. Explain the concept of a discriminator (partial key) in a weak entity.
7. Explain one-to-many cardinality with an example.
8. Explain many-to-many cardinality with an example.
9. What is total participation and partial participation? Give examples of each.
10. Explain generalization and specialization with an example.
11. Explain aggregation in the ER model with an example.
12. Explain the rule for converting a strong entity set into a relational table.
13. Explain the rule for converting a weak entity set into a relational table.
14. Explain how a 1:1 relationship is mapped into relational tables.
15. Explain how a many-to-many (M:N) relationship is converted into a relational table.
16. Explain functional dependency with an example: StudentID → Name.
17. State Armstrong's Axioms (Reflexivity, Augmentation, Transitivity) briefly.
18. What is the role of a candidate key in normalization?
19. Differentiate between partial and full functional dependency.
20. Explain lossless join decomposition and why it is important.

Section C

1. Draw an ER diagram for a "College Management System" with entities STUDENT, COURSE, and DEPARTMENT, appropriate relationships, cardinalities and participation constraints, and convert the ER diagram into a relational schema (tables with primary/foreign keys).
2. Explain the rules for converting an ER diagram to relational tables, covering strong entities, weak entities, 1:1, 1:N, and M:N relationships, with suitable examples.
3. Explain the Extended ER features — Generalization, Specialization, and Aggregation — with suitable diagrams and examples.
4. Explain functional dependency theory including Armstrong's Axioms and their use in computing the closure of a set of attributes, with an example.
5. Explain the process of normalization up to BCNF with a suitable un-normalized relation as an example, showing the schema at each normal form.
6. What is a multivalued dependency? Explain Fourth Normal Form (4NF) with a suitable example relation.
7. Explain join dependency and Fifth Normal Form (5NF) with an example.
8. Explain the concept of dependency preservation in normalization, illustrating with an example a decomposition that is and one that is not dependency-preserving.

9. Design an ER diagram for an "Online Banking System" with entities such as CUSTOMER, ACCOUNT, BRANCH, and TRANSACTION, showing relationships and constraints, and convert it into a normalized relational schema.
10. Explain the removal of redundant attributes in entity sets and the reduction of an ER diagram to a relational schema, with an example covering composite and multivalued attributes.